

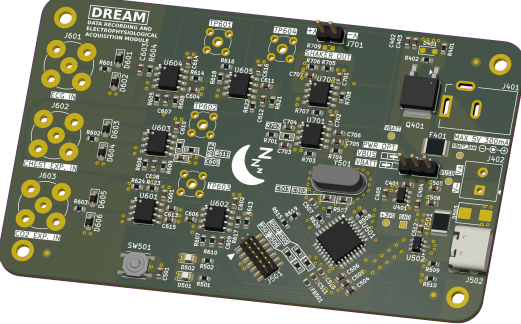
DREAM

VARIANT: RELEASED

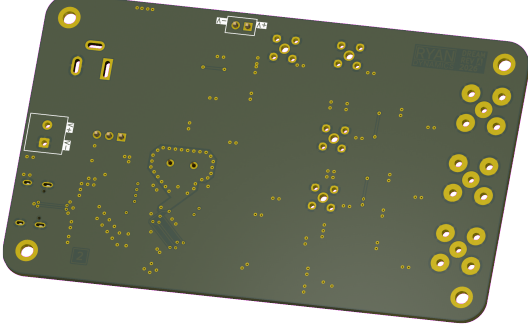
2026-05-09
REV 1.2.0

Page	Index	Page	Index	Page	Index	Page	Index
1	COVER PAGE	11		21		31	
2	CIRCUIT	12		22		32	
3	REVISION HISTORY	13		23		33	
4		14		24		34	
5		15		25		35	
6		16		26		36	
7		17		27		37	
8		18		28		38	
9		19		29		39	
10		20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

"Data Recording and Electrophysiological Acquisition Module"

NOT FITTED COMPONENTS MARKED AS **X**

DRAFT – EARLY STAGE SCHEMATIC CAPTURE
PRELIMINARY – LATE STAGE SCHEMATIC CAPTURE
CHECKED – PCB LAYOUT CHECKED, CONTACT ENGINEER IF MISTAKE FOUND
RELEASED – READY FOR PRODUCTION

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

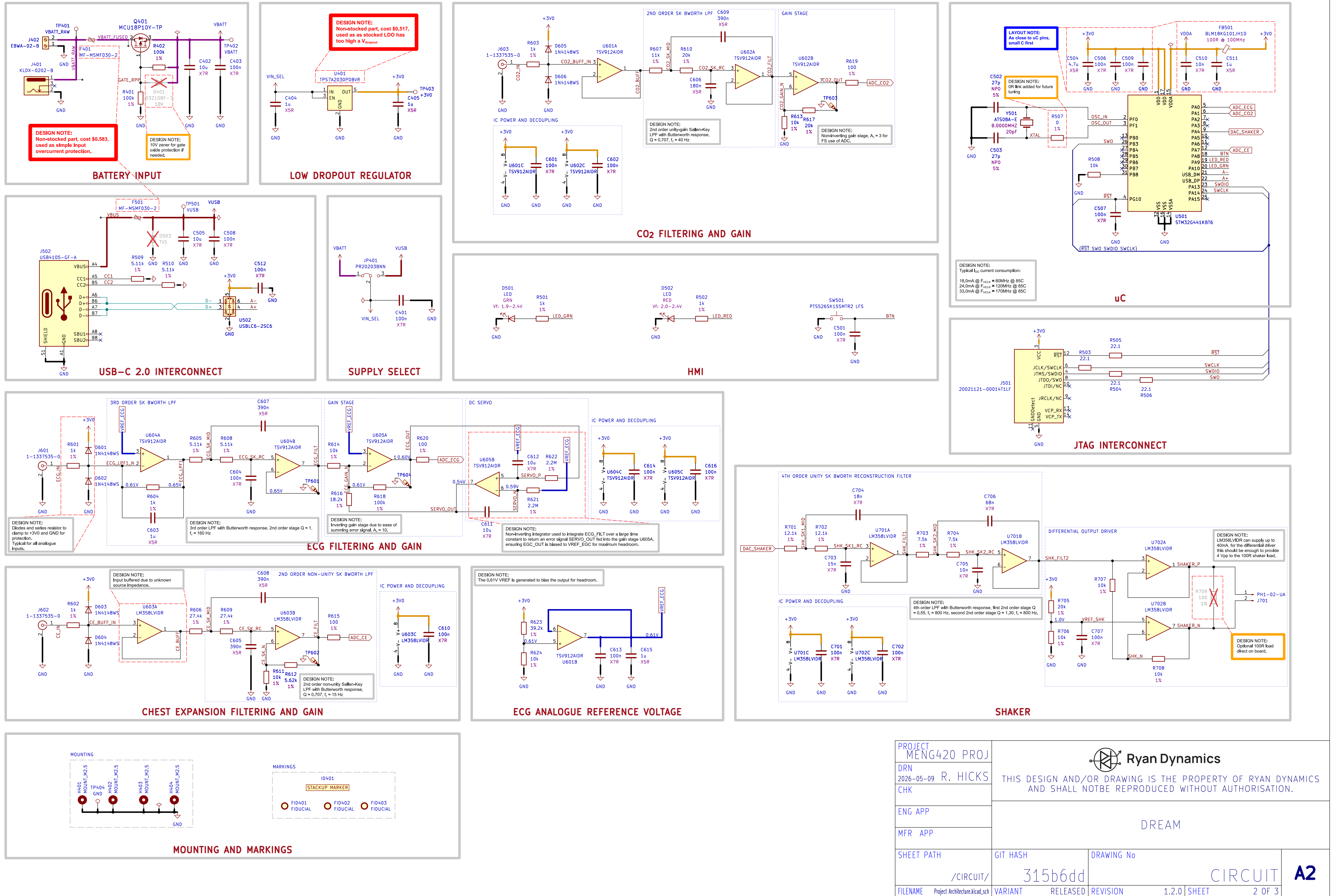
DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

PROJECT MENG420 PROJ	 Ryan Dynamics THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.		
DRN 2026-05-09 R. HICKS			
CHK			
ENG APP	DREAM		
MFR APP			
SHEET PATH	GIT HASH 315b6dd	DRAWING No	A3
FILENAME DREAM.kicad_sch	VARIANT RELEASED	REVISION 1.2.0	
		SHEET 1 OF 3	

[2] CIRCUIT



[3] REVISION HISTORY

Version 1.0.0 - 2026-04-14

Added

- Schematic capture
- Preliminary PCB layout

Version 1.1.0 - 2026-04-14

Fixed

- Add ground pour underneath uC

Version 1.2.0 - 2026-05-09

Added

- Added 3D models to some footprints missing them

Fixed

- Changed gain of ECG chain to x10 from x24.9
- Compiled entire schematic into single sheet

Version 1.0.1 - 2026-04-14

Added

- Ground pour underneath XTAL.
- Untented via flag set
- Production files

Version 1.0.2 - 2026-04-14

Added

- Added some silkscreen flair
- Broke out unused STM32 pins to vias
- Added pin 1 indicator to JTAG header

Fixed

- Changed VBUS_RAW to VBUS
- Changed VBUS to VUSB

Version 1.0.3 - 2026-04-14

Changed

- Crystal oscillator layout, isolated ground pour, ground perimeter

PROJECT
MENG420 PROJ

DRN
2026-05-09 R. HICKS

CHK

ENG APP

MFR APP

SHEET PATH

/REVISION HISTORY/

FILENAME Revision History.kicad_sch



Ryan Dynamics

THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS
AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.

DREAM

GIT HASH

315b6dd

VARIANT

RELEASED

DRAWING No

REVISION HISTORY

REVISION

1.2.0

SHEET

3 OF 3

A4